- A *set* is an unordered collection of "objects".
- The *objects* in a set are called the elements or members of the set.
- A set is usually denoted by a capital letter, say A, and an element is usually denoted by a lowercase letter, say a
- We write $a \in A$ to mean "a is an element of A".
- $a \notin A$ means the statement $\sim(a \in A)$, or "a is not an element of A".

- A set is determined by a defining property P of its elements, written as $\{x : P(x)\}$
- To avoid running into logical difficulties we shall always assume that our objects which are described by the predicate P come from a previously well defined set. say M, and sometimes we shall say so explicitly. In general then, we describe sets by

$$\{x : x \in M \text{ and } P(x)\},$$

which also can be written as $\{x \in M : P(x)\}$.

3.1 Subset, Equality of sets, singleton set, The empty set, Universal set.

- The set A is a **subset** of the set B if and only if every element of A is also an element of B, and denote this as $A \subseteq B$.

Note:

- $A \subseteq B$ if and only if $\forall x, x \in A \implies x \in B$.

- A set A is a *proper subset* of a set B if $(\forall x, x \in A \Rightarrow x \in B) \wedge (\exists x_0, x_0 \in B \wedge x_0 \notin A)$.

- Two sets A and B are ***equal*** if and only if A $\subseteq$ B and B $\subseteq$ A.

  This can be interpreted as A = B if and only if $\forall x, x \in A \Leftrightarrow x \in B$.

- A set containing exactly one element is called a *singleton set*.

- The set with no elements is called the *empty* set, denoted $\emptyset$ or $\{ \}$,

  i.e $A = \phi \iff \sim(\exists x, x \in A) \equiv \forall x, x \notin A$.

PMT 1013 Foundations of Mathematics

Mid- Exam 01

Date: 08.06. 2026 **Time:** 01 hour

Name with initials: **Index No:**

Answer **all** questions in the given space.

Definition: Suppose A is a set. Then the *power set of A* is denoted by $\mathcal{P}(A)$ and is given by, $\mathcal{P}(A) = \{X \mid X \text{ is a set and } X \subseteq A\}$

For example, let $A = \{0, 1\}$

Then $\mathcal{P}(A) = \{\phi, \{0\}, \{1\}, \{0,1\}\}$

Definition: In any application of set theory the members of the sets that are considered usually belong to some largest set called the *universal set.* This set is usually denoted by the letter U or E. For example, if we are only considering subset of $\mathbb{R}$, the universal set will be $\mathbb{R}$.

3.2 Set Operations

Definition: Let A and B be sets. The union of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or B or both. That is,

$$A \cup B = \{x : x \in A \vee x \in B\}$$

Definition: Let A and B be sets. The intersection of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B. That is,

$$A \cap B = \{x : x \in A \wedge x \in B\}$$

Definition : Suppose A is a subset of a universal set E. Then the complement of A is denoted by A^c and is given by

$$A^c = \{x | x \in E \wedge x \notin A\} = A'$$

Definition : Let A and B be sets. The difference of A and B, denoted by $A \setminus B$, is the set containing those elements that are in A but not in B. That is,

$$A \setminus B = \{x : x \in A \wedge x \notin B\}$$

$A \setminus B$ is also called the complement of A in B or the complement of A with respect to B.

Ex: Prove that:

non-empty

(i) "If **Set A and Set B are two sets such that $A \subseteq B$ then $A \cup B = B$**"

(ii) If **Set A and Set B are two sets such that $A \subseteq B$ then $A \cap B = A$**

Ex: Let A and B be two non-empty subsets of the universal set E.
Prove or disprove the following.
(i) If $A \subseteq B$, then $A \cap B = B$ (ii) If $A \subseteq B$, then $A \cup C \subseteq B \cap C$ for any set C.

3.3 Set Identities

Identity Laws

$A \cup \emptyset = A$

$A \cap U = A$

Commutative Laws

$A \cup B = B \cup A$

$A \cap B = B \cap A$

Domination Laws

$A \cup U = U$

$A \cap \emptyset = \emptyset$

Associative Laws

$A \cup (B \cup C) = (A \cup B) \cup C = A \cup B \cup C$

$A \cap (B \cap C) = (A \cap B) \cap C = A \cap B \cap C$

De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Absorption Laws

$A \cup (A \cap B) = A$

$A \cap (A \cup B) = A$

Idempotent Laws

$A \cup A = A$

$A \cap A = A$

Distributive Laws

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Complement Laws

$$A \cup A^c = U$$

$$A \cap A^c = \emptyset$$

Complementation Law

$$(A^c)^c = A$$

End of D_{14}